

Microcontroller and Microprocessor Project

Laboratory report submitted for the partial fulfillment of the
requirements for the degree of

**Bachelor of Technology in
Electronics and Communication Engineering**

RFID BASED ATTENDANCE SYSTEM

by

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ABSTRACT

This project presents a simple, practical, and efficient RFID-based attendance system designed to overcome the limitations of traditional manual attendance methods. In this system, each student is assigned a unique RFID card, which is scanned using an RC522 reader connected to an Arduino microcontroller. When a card is tapped, the system reads the UID, verifies it with the stored database, and immediately displays the student's name on a 16×2 LCD while giving a buzzer confirmation. The attendance is recorded automatically, reducing time, eliminating manual errors, and ensuring a smooth classroom workflow. The overall design is user-friendly, easy to install, and reliable for day-to-day use. The project demonstrates how RFID technology can provide a faster, cleaner, and more organized approach to attendance management while also improving accuracy and convenience. This system can be implemented in schools, colleges, and institutions where quick, error-free attendance tracking is essential.

Acknowledgment

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With Regards

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INTRODUCTION

Attendance tracking is an essential part of academic and organizational management, ensuring that records are accurate, reliable, and easy to maintain. Traditional manual attendance methods, such as roll-calling or maintaining registers, are time-consuming and prone to human error. With the increasing need for automation and accuracy, technology-driven attendance systems have gained significant importance. Among these technologies, Radio Frequency Identification (RFID) has emerged as a fast, secure, and efficient solution.

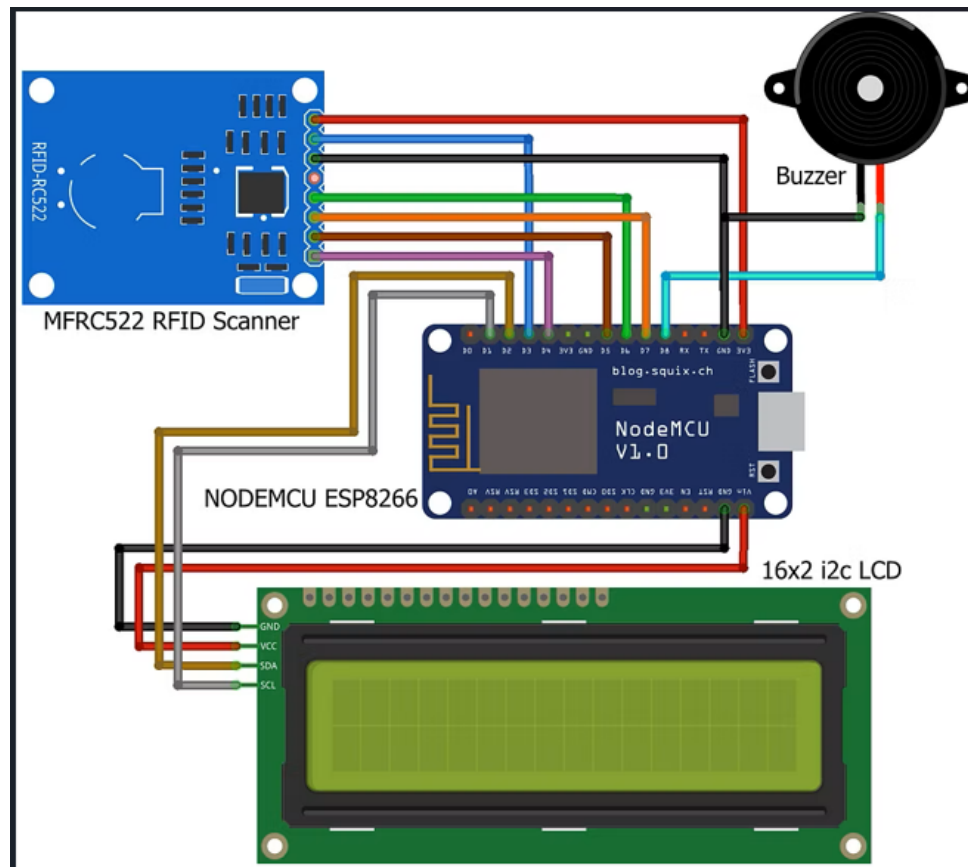
RFID is a wireless identification technology that allows information to be transmitted without physical contact. It uses RFID tags or cards that store a unique identification number (UID), which can be read by an RFID reader when brought into proximity. This contactless feature makes RFID one of the most suitable methods for automated attendance systems, especially in environments where speed and accuracy are crucial.

In an RFID-based attendance system, each user is issued an RFID card encoded with a unique identifier. When the card is tapped on the reader, the system reads the UID, verifies it with the database, and instantly logs the attendance. This process eliminates manual entry, reduces paperwork, and minimizes the chances of proxy or incorrect attendance marking.

The integration of microcontrollers, RFID modules, and display units further enhances the system's functionality. Microcontrollers such as Arduino or ESP8266 process the RFID data and communicate with external components, while LCD displays show user details for confirmation. Additional features like buzzers or LEDs provide immediate feedback to ensure a smooth user experience.

Overall, this project demonstrates how RFID technology can transform conventional attendance systems into efficient, error-free, and user-friendly solutions. By automating the attendance process, institutions can improve accuracy, save time, and maintain well-organized records suitable for both academic and professional environments.

Circuit Diagram



SWOT ANALYSIS

SWOT Analysis of RFID-Based Attendance System	
Strengths	Weaknesses
• Fast and contactless attendance marking. • Reduces human error and proxy attendance. • User-friendly and easy to operate. • Real-time data logging with accurate timestamps.	• Requires hardware like RFID module and microcontroller. • Limited range; user must tap near the reader. • System depends on power and internet connectivity (if cloud-based). • Cards can be damaged or misplaced.
Opportunities	Threats
• Can be expanded for hostel entry, library, labs, and offices. • Can integrate with mobile apps or cloud dashboards. • Future upgrade: biometric + RFID hybrid system. • Useful for large institutions needing fast attendance.	• RFID cloning risk if security is not implemented. • Hardware failure can stop attendance marking. • Multiple tags near the reader may cause read errors. • Environmental interference may affect signal detection.

Chapter 1

RFID BASED ATTENDANCE SYSTEM

1.1. Aim

To design and implement an RFID-based attendance system that automatically records and stores the attendance of individuals using RFID cards.

1.2. Apparatus Used

- ESP8266(NODEMCU)
- RFID reader (RC522)
- RFID tags/cards
- 16x2 LCD display
- Buzzer + LEDs
- Jumper wires + breadboard

1.3. Software Used

- Arduino ide

1.4. Theory

Radio Frequency Identification (RFID) is a wireless identification technology that uses radio waves to automatically identify and track tags attached to objects or individuals. An RFID-based attendance system uses this technology to record attendance without manual entry, improving accuracy and efficiency.

Key Features

- Contactless Identification – Attendance is marked simply by bringing the RFID card near the reader, enabling quick and hygienic operation.
- Unique User Authentication – Each RFID tag has a unique ID, ensuring accurate identification and preventing proxy attendance.
- Real-Time Data Recording – Attendance is logged instantly along with date and time using a microcontroller or RTC module.
- High Speed Efficiency – The system reads RFID tags within milliseconds, allowing fast entry of multiple users
- Integration Capability – Can be integrated with displays, alarms, cloud storage, or Google Sheets for automated attendance tracking.

1.6 Hardware Connections

RTC Module (DS1307):

- SCL → P0.6
- SDA → P0.7

LCD Connections:

- D0–D7 → Port 2
- RS → P0.0
- RW → P0.1
- E → P0.2

Switches:

- D1 (Hour +) $\rightarrow$ P1.4
- D2 (Min +) $\rightarrow$ P1.5
- D3 (Sec +) $\rightarrow$ P1.6
- D4 (Confirm Alarm) $\rightarrow$ P1.7
- D5 (Stop Alarm) $\rightarrow$ P0.3

Output:

- Buzzer $\rightarrow$ P1.1
- LED $\rightarrow$ P1.0

1.5. Program / Code

```
// Include LCD library for I2C-based 16x2 LCD
#include <LiquidCrystal_I2C.h>

// Viral Science project credits (original code author)
#include <SPI.h> // SPI communication library for RFID
#include <MFRC522.h> // RFID module library
#include <Arduino.h> // Arduino core functions
#include <ESP8266WiFi.h> // ESP8266 WiFi handling
#include <ESP8266HTTPClient.h> // HTTP GET/POST handling
#include <WiFiClient.h> // WiFi client
#include <WiFiClientSecureBearSSL.h> // Secure HTTPS client
#include <LiquidCrystal_I2C.h> // LCD library (included twice but harmless)

//-----
// Pin definitions for NodeMCU + RC522 module
#define RST_PIN D3
#define SS_PIN D4
#define BUZZER D8
//-----

// Create RFID module object
MFRC522 mfrc522(SS_PIN, RST_PIN);

// Create objects for authentication keys
MFRC522::MIFARE_Key key;
MFRC522::StatusCode status;

//-----
// Memory block number on RFID card to read data from
int blockNum = 2;

// Length of buffer to store 16 bytes + 2 extra bytes
byte bufferLen = 18;
byte readBlockData[18]; // Memory to store block data

//-----
// String to store student name read from RFID card
String card_holder_name;

// Google Sheet Script URL with name parameter
```

```

const String sheet_url =
"https://script.google.com/macros/s/
    AKfycbzoex8T2HDkaMnu3kfz4WC2tCHpBelS8a5xRBNGsoNQ7UqvCvtZH4mdpk__IE1jgk1I/
    exec?name=";

//-----
// WiFi credentials
#define WIFI_SSID "Rfid"
#define WIFI_PASSWORD "nlbi7410"

//-----
// Initialize LCD: Address 0x3F, Display size 16x2
LiquidCrystal_I2C lcd(0x3F, 16, 2);

/*
*****

* setup() function
*****

*/
void setup()
{
    // Initialize Serial Monitor
    Serial.begin(9600);

    // Initialize LCD
    lcd.begin(16,2);
    lcd.backlight();
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("  Initializing  ");

    // Display loading animation
    for (int a = 5; a <= 10; a++) {
        lcd.setCursor(a, 1);
        lcd.print(".");
        delay(500);
    }

    //-----
    // Connect to WiFi
    Serial.println();
    Serial.print("Connecting to AP");

```

```

WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

// Loop until WiFi is connected
while (WiFi.status() != WL_CONNECTED){
  Serial.print(".");
  delay(200);
}

Serial.println("");
Serial.println("WiFi_connected.");
Serial.println("IP_address:");
Serial.println(WiFi.localIP());

//-----
// BUZZER pin output
pinMode(BUZZER, OUTPUT);

//-----
// Start SPI communication (needed for RC522)
SPI.begin();
}

/*
*****

* loop() function  Runs repeatedly
*****

*/
void loop()
{
  // Display message on LCD asking to scan card
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("_Scan_your_Card_");

  // Initialize RFID module each loop
  mfrc522.PCD_Init();

  // Look for a new RFID card
  if (!mfrc522.PICC_IsNewCardPresent()) {return;}

  // Select the detected RFID card
  if (!mfrc522.PICC_ReadCardSerial()) {return;}

```

```

//-----
// Read data block from card
Serial.println();
Serial.println(F("Reading data from RFID..."));

ReadDataFromBlock(blockNum, readBlockData);

Serial.println();
Serial.print(F("Last data in RFID block"));
Serial.print(blockNum);
Serial.print(F(" -->"));

// Display the name on LCD and Serial
for (int j=0 ; j<16 ; j++)
{
    Serial.write(readBlockData[j]);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Hey" + String((char*)readBlockData) + "!");
}

Serial.println();

//-----
// Beep Buzzer twice on successful scan
digitalWrite(BUZZER, HIGH); delay(200);
digitalWrite(BUZZER, LOW); delay(200);
digitalWrite(BUZZER, HIGH); delay(200);
digitalWrite(BUZZER, LOW);

//-----
// Send attendance data to Google Sheets (HTTPS)
if (WiFi.status() == WL_CONNECTED) {

    // Create secure client for HTTPS
    std::unique_ptr<BearSSL::WiFiClientSecure> client(new BearSSL::
        WiFiClientSecure);

    // Ignore SSL certificate (NOT recommended for real systems)
    client->setInsecure();
}

```



```

// Build URL: script_url + name_from_card
card_holder_name = sheet_url + String((char*)readBlockData);
card_holder_name.trim();
Serial.println(card_holder_name);

HTTPClient https;
Serial.print(F("[HTTPS]_begin...\n"));

if (https.begin(*client, (String)card_holder_name)) {

    Serial.print(F("[HTTPS]_GET...\n"));
    int httpCode = https.GET(); // Perform GET request

    if (httpCode > 0) {
        Serial.printf("[HTTPS]_GET..._code:_%d\n", httpCode);

        // Show success message on LCD
        lcd.setCursor(0, 1);
        lcd.print("_Data_Recorded_");
        delay(2000);
    }
    else {
        Serial.printf("[HTTPS]_GET_failed:_%s\n",
                      https.errorToString(httpCode).c_str());
    }

    https.end();
    delay(1000);
}
else {
    Serial.printf("[HTTPS]_Unable_to_connect\n");
}
}

/*
*****

* Function: ReadDataFromBlock()
* Purpose: Authenticates and reads a block of data from RFID card
*****

*/
void ReadDataFromBlock(int blockNum, byte readBlockData[])

```

```

{
    // Load default key = 0xFFFFFFFFFFFF
    for (byte i = 0; i < 6; i++) {
        key.keyByte[i] = 0xFF;
    }

    // Authenticate block for reading using Key A
    status = mfrc522.PCD_Authenticate(
        MFRC522::PICC_CMD_MF_AUTH_KEY_A,
        blockNum,
        &key,
        &(mfrc522.uid));

    if (status != MFRC522::STATUS_OK){
        Serial.print("Authentication failed: ");
        Serial.println(mfrc522.GetStatusCodeName(status));
        return;
    }
    else {
        Serial.println("Authentication success");
    }

    // Read 16 bytes from block
    status = mfrc522.MIFARE_Read(blockNum, readBlockData, &bufferLen);

    if (status != MFRC522::STATUS_OK) {
        Serial.print("Reading failed: ");
        Serial.println(mfrc522.GetStatusCodeName(status));
        return;
    }
    else {
        Serial.println("Block was read successfully");
    }
}

```

(→ Continue your entire program here, paste full code)

1.8 Program / Code Structure

1. LCD Functions:

- `lcd_cmd()`: Sends a command to the LCD.
- `lcd_data()`: Sends data to the LCD.
- `lcd_init()`: Initializes the LCD for 2-line display and clears the screen.

2. I2C Communication:

- `I2C_Start()`: Starts communication with the RTC module.
- `I2C_Stop()`: Ends communication with the RTC module.
- `I2C_Write()`: Writes data to the RTC module.
- `I2C_Read()`: Reads data from the RTC module.
- `I2C_Ack()` and `I2C_NoAck()`: Acknowledge signals during data transfer.

3. Alarm Functions:

- `set_alarm()`: Allows the user to set an alarm using switches.
- `display_set_alarm()`: Displays the currently set alarm on the LCD.

4. Delay Functions:

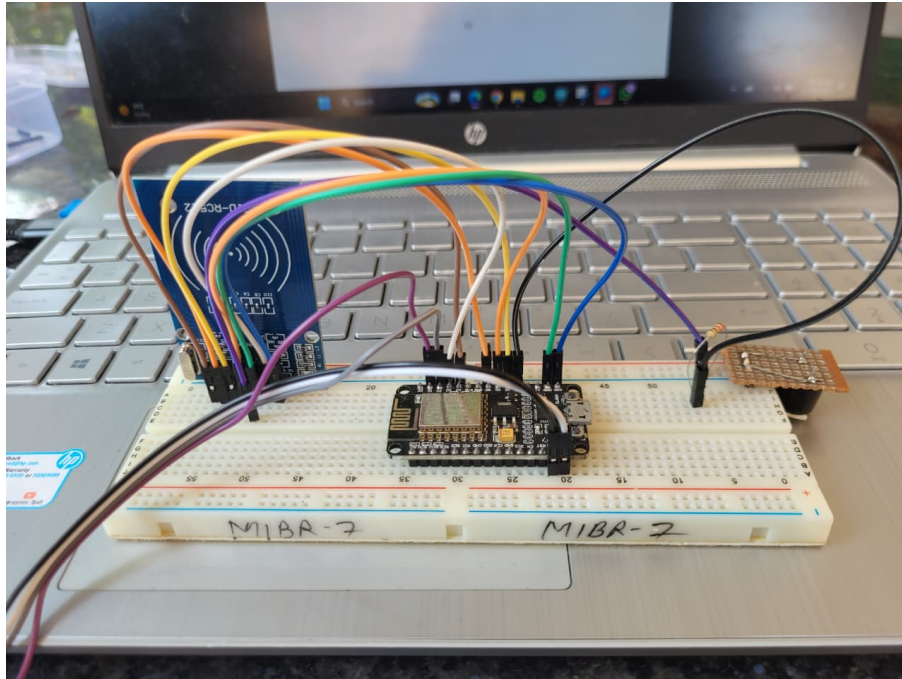
- `delay_us()`, `delay_ms()`, `delay_sec()`: Provide microsecond, millisecond, and second delays, respectively.

5. Main Routine:

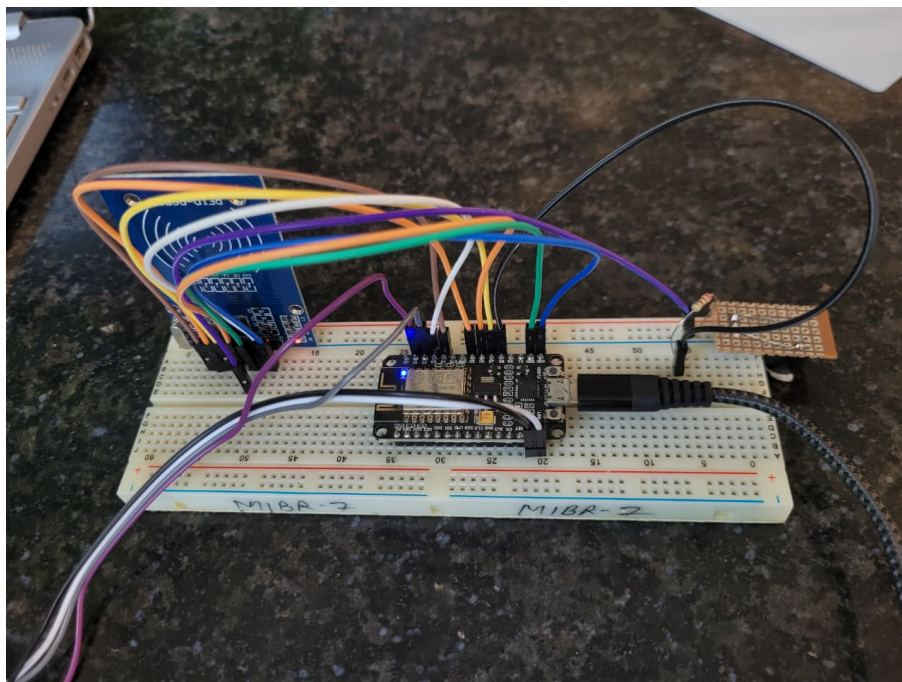
- Initializes the LCD and RTC.
- Continuously reads the time and date from the RTC and displays them.
- Triggers the alarm when the set alarm time matches the current time.

1.6. Observations on Hardware

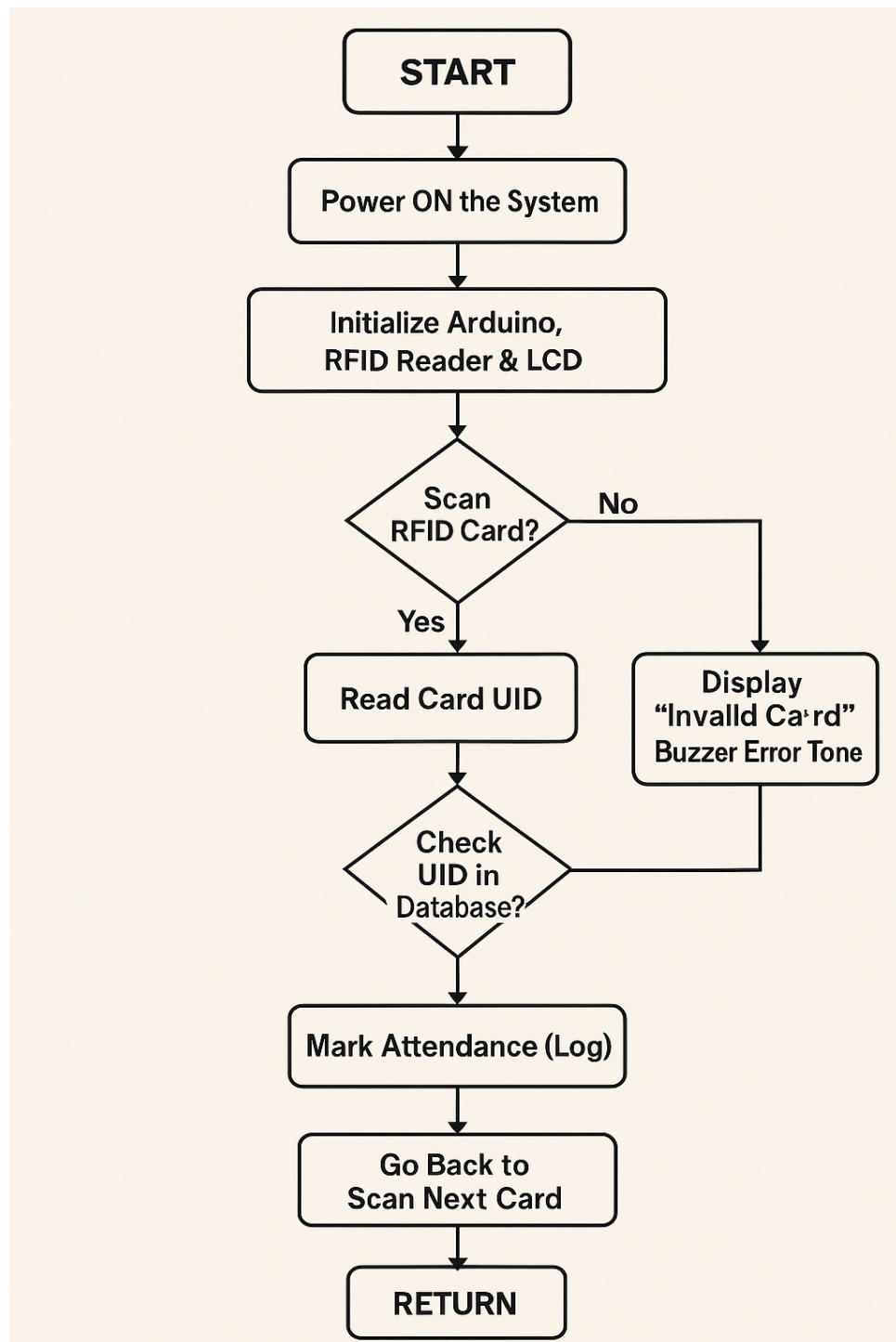
CIRCUIT WHEN WIFI IS NOT CONNECTED :



CIRCUIT WHEN WIFI IS CONNECTED :



1.7. Flow Chart



1.8. Conclusion

The RFID-based Attendance System successfully demonstrates the use of Radio Frequency Identification technology for automating attendance recording in educational institutions, offices, and secured environments. By using RFID tags and a reader module, the system ensures fast, contactless, and reliable identification of individuals, eliminating the need for manual attendance registers and reducing human errors. Each RFID tag carries a unique ID, allowing the system to accurately recognize users and prevent issues like proxy attendance.

This project demonstrates key concepts of embedded systems, serial communication, and automation. It highlights the advantages of RFID over traditional methods in terms of speed, accuracy, and security. Overall, the RFID-based Attendance System is a cost-effective and scalable solution that can be implemented in various real-world applications where reliable attendance tracking is essential. The project can be further enhanced by adding mobile notifications, database integration, or biometric verification for additional security.

1.9. Precautions

1. Ensure all RFID reader and microcontroller connections are made properly and securely to avoid malfunction or incorrect attendance detection.
2. Keep RFID tags away from strong magnetic fields or physical damage, as this may affect their readability and performance.

1.10. Bibliography

Viral Science, “RFID Based Attendance System using Arduino and Google Sheets,” YouTube, 2023. [Online]. Available: <https://www.youtube.com/>

D. M. Dobkin, *The RF in RFID: Passive UHF RFID in Practice*. Burlington, MA, USA: Elsevier, 2012.

S. R. Bharamagoudar, R. B. Geeta, and S. G. Totad, “Web-Based Student Attendance System Using RFID,” *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 3, no. 1, pp. 39–44, 2013.

NXP Semiconductors, “MFRC522 Standard High Performance MIFARE Reader IC Datasheet,” 2019. [Online]. Available: <https://www.nxp.com>

Arduino, “Arduino Uno Technical Specifications,” 2024. [Online]. Available: <https://www.arduino.cc>

Practical Engineering, “How RFID Works,” YouTube, 2022. [Online]. Available: <https://www.youtube.com/>

GreatScott!, “Using the RC522 RFID Reader with Arduino,” YouTube, 2021. [Online]. Available: <https://www.youtube.com/>